

KLERIMED – PACKAGE INSERT

1. Trade Name of the Medicinal Product

KLERIMED 250MG TABLET

KLERIMED 500MG TABLET

2. Qualitative and Quantitative Composition

Each KLERIMED tablet contains 250mg or 500mg of clarithromycin

3. Pharmaceutical Form

Film coated tablets for oral administration.

4. Clinical Particulars

4.1. Therapeutic Indications

For the treatment of infections caused by one or more susceptible organisms. The indications include:

- initial therapy in community acquired respiratory infections
- upper respiratory tract infection, including pharyngitis and sinusitis
- lower respiratory tract infection, including acute and chronic bronchitis and pneumonia
- skin and soft tissue infections of mild to moderate seriousness
- eradication of *Helicobacter pylori* in duodenal ulcer patients when used in the presence of acid suppression induced by omeprazole. The activity is greater at neutral pH than at acid pH.

Clarithromycin is usually active, *in vitro*, against the following organisms:

Gram-positive bacteria: *Staphylococcus* (methicillin susceptible), *Streptococcus pyogenes* (Group A β haemolytic streptococci), α haemolytic streptococci (viridans group), *Streptococcus pneumoniae*, *Streptococcus agalactiae*, *Listeria monocytogenes*

Gram-negative bacteria: *Bordetella pertussis*, *Campylobacter jejuni*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Helicobacter pylori*, *Legionella pneumophila*, *Moraxella catarrhalis* and *Neisseria gonorrhoeae*.

Mycoplasma: *Mycoplasma pneumoniae*, *Ureaplasma urealyticum*.

Anaerobes: *Bacteroides fragilis* (macrolide susceptible), *Clostridium perfringens*, *Peptococcus sp.*, *Peptostreptococcus sp.*, *Propionibacterium acnes*.

Other organisms: *Chlamydia trachomatis*, *Mycobacterium avium*, *Mycobacterium chelonae*, *Mycobacterium fortuitum*, *Mycobacterium intracellulare*, *Mycobacterium kansasii* *Mycobacterium leprae*.

The activity is bactericidal against several bacterial strains, including *Campylobacter sp.*, *Haemophilus influenzae*, *Helicobacter pylori*, *Moraxella catarrhalis*, *Neisseria gonorrhoeae*, *Streptococcus agalactiae*, *Streptococcus pneumoniae* and *Streptococcus pyogenes*.

4.2. Posology and Method of Administration

For oral administration. Food does not affect the bioavailability.

Adults: the usual dose is 250mg two times a day for seven days. In severe infections this may be increased to 500mg two times a day for fourteen days.

H. pylori eradication in duodenal ulcer patients (Adults): alternative therapies are effective:

Triple therapy (7 days)	Triple therapy (10 days)	Dual therapy (14 days)
500mg clarithromycin two times a day 40mg omeprazole once a day 1000mg amoxicillin two times a day	500mg clarithromycin two times a day 20mg omeprazole once a day 1000mg amoxicillin two times a day	500mg clarithromycin three times a day 40mg omeprazole once a day

Children: children of twelve (12) years or older, as the adult dosage.

Children younger than twelve (12) years should use the oral suspension only.

Elderly: as for adults.

Renal impairment: usually dosage reduction is not required unless there is severe renal impairment, creatinine clearance <30ml/min. In such cases the total daily dose should be halved to 250mg once a day or 250mg two times a day in severe infections.

Hepatic impairment: usually dosage reduction is not required, but as the liver is a principal route of excretion caution should be exercised.

4.3. Contra-indications

Patients with known hypersensitivity to macrolide antibiotics.

Concomitant administration with ergot derivatives.

Concomitant administration with any of the following: cisapride, pimozide, terfenadine. Elevated levels of those drugs have been reported in patients concomitantly administered clarithromycin. This may result in QT prolongation and cardiac arrhythmias including ventricular fibrillation, ventricular tachycardia and Torsade de Pointes. Similar effects have been observed with concomitant administration of astemizole and other macrolides.

Concomitant administration of clarithromycin and the following drugs is contraindicated: Domperidone as this may result in QT prolongation and cardiac arrhythmias including ventricular tachycardia, ventricular fibrillation, and torsades de pointes (*See Interactions*)

4.4. Special Warnings and Precautions for Use

As clarithromycin is principally excreted by the liver and kidney, caution should be exercised in administration to patients with either impaired hepatic function or impaired renal function. Super-infection due to the overgrowth of non-susceptible bacteria or fungi may occur on prolonged or repeated use of clarithromycin. If it occurs, clarithromycin should be discontinued and appropriate therapy instituted.

In a small number of patients, *H. pylori* may develop resistance to clarithromycin.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], KLERIMED should be discontinued immediately and appropriate treatment should be urgently initiated.

4.5. Interactions with other Medicaments and other forms of Interaction

Non-interaction with oral contraceptives has been shown.

In common with other macrolide antibiotics, patients concurrently taking drugs that are metabolised by the cytochrome P-450 system, such as cyclosporin, disopyramide, ergot alkaloids, lovastatin, midazolam, phenytoin, triazolam, warfarin, may have an elevation of serum levels of those drugs.

Administration of clarithromycin to patients on theophylline therapy has been associated with elevated serum levels of theophylline and potential theophylline toxicity.

Administration of clarithromycin to patients on warfarin therapy may result in potentiation of warfarin effects and prothrombin time should be frequently monitored.

Concomitant administration of clarithromycin and digoxin may result in potentiation of digoxin effects and the monitoring of serum digoxin levels should be considered.

The effects of carbamazepine may be potentiated by clarithromycin due to a reduction in the rate of carbamazepine excretion.

Clarithromycin and zidovudine in HIV infected adults may result in a decrease of zidovudine steady state levels, which can be largely avoided by staggering the doses of each by one to two hours. This effect has not been reported in children.

Increased plasma concentrations of clarithromycin and omeprazole may result on concurrent administration, dosage reduction is unnecessary. Clarithromycin plasma levels may also be increased with concurrent administration of ranitidine, similarly no dosage reduction is required.

Co-administration of clarithromycin, known to inhibit CYP3A, and a drug primarily metabolised by CYP3A may be associated with elevations in drug concentrations that could increase or prolong both therapeutic and adverse effects of the concomitant drug.

The following drugs or drug classes are known or suspected to be metabolised by CYP3A isozyme:

Domperidone.

4.6. Pregnancy and Lactation

Safety of clarithromycin in pregnancy and breast feeding is not established. Clarithromycin should not be used in pregnancy unless the benefit is considered to outweigh the potential risk to the foetus. Some animal studies have suggested an embryotoxic effect, although only at dose levels that were clearly toxic to the mother also.

If administration of clarithromycin is necessary in breast feeding, breast feeding should be discontinued. Clarithromycin has been found in milk of lactating animals and humans.

4.7. Effects on Ability to Drive and Use Machines

No known effects

4.8. Undesirable Effects

In general, clarithromycin is well tolerated. The following side effects have been reported:

Gastro-intestinal: nausea, dyspepsia, abdominal pain, diarrhoea and vomiting. Stomatitis and glossitis have been reported.

Skin: allergic reactions have been reported, ranging from urticaria, mild skin eruptions to anaphylaxis rarely Steven's-Johnson syndrome, or toxic epidermal necrolysis. In clinical trials with concomitant administration of clarithromycin and omeprazole, reversible tongue discolouration and oral monilia has been seen.

Central nervous system: headache. Other transient effects are anxiety, dizziness, insomnia, confusion, nightmares, hallucinations and psychoses, although cause and effect has not been established, vertigo, tinnitus and depersonalization. Hearing loss has occurred this is usually reversible upon discontinuation of therapy. Taste perversion may occur, usually in conjunction with alteration of the sense of smell. There have been reports of tooth discoloration, but it is usually reversible with professional dental cleaning.

Hepatic system: hepatic dysfunction, as with other macrolides, has been reported. This is usually reversible, and ranges from altered liver function tests, hepatitis and cholestasis, with or without jaundice. The dysfunction may be severe, and fatal hepatic dysfunction has been reported very rarely.

Skin and Subcutaneous Tissue Disorders:

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)

Other: Arthralgia, and myalgia has been reported. Oral candidiasis has been reported. Rarely pseudomembranous colitis may occur, ranging in severity from mild to life threatening. Cases of increased serum creatinine interstitial nephritis, renal failure, pancreatitis and convulsions have been reported rarely. As with other macrolides, QT prolongation, ventricular tachycardia and Torsade de pointes have been rarely reported. There have been rare reports hypoglycaemia, some of which have occurred in patients on concomitant oral hypoglycaemic agents or insulin. There have been very rare reports of ureitis mainly in patients treated with concomitant rifabutin, most of these were reversible. Isolated cases of leukopenia and thrombocytopenia have been reported.

4.9. Overdose

Ingestion of large amounts of clarithromycin would be expected to produce gastro-intestinal symptoms, nausea and vomiting and diarrhoea. A report exists of the ingestion of 8g of clarithromycin by a patient with a history of bipolar disorder, who exhibited altered mental status, paranoid behaviour and hypokalemia and hypoxemia. Allergic reactions initiated by overdose should be treated with gastric lavage and supportively. Haemodialysis or peritoneal dialysis, as with other macrolides, would not be anticipated to help.

5. Pharmacological Properties

5.1. Pharmacodynamic Properties

Clarithromycin is a semi-synthetic derivative of erythromycin A. It acts in the same way, binding to the 50s ribosomal sub-unit and suppressing protein synthesis in susceptible bacteria. The 14-hydroxy metabolite also has anti-bacterial activity, although the MIC (minimum inhibitory concentration) is either equal or two times higher than the MIC of the parent compound, except in the case of *H. influenzae*, where it is two times more active than clarithromycin. The MICs of clarithromycin are generally two times lower than for erythromycin.

Clarithromycin is usually active, *in vitro*, against the following organisms:

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5.2. Pharmacokinetic Properties

Clarithromycin is rapidly and well absorbed from the gastro-intestinal tract following oral administration. First pass metabolism is responsible for the formation of the microbiologically active 14-hydroxycarithromycin metabolite. Food does not affect bioavailability, although it slightly delays absorption of clarithromycin and the 14-hydroxymetabolite formation.

Clarithromycin has non-linear pharmacokinetics, but steady state is reached within two days of dosing. Doses of 250mg two times a day result in urinary excretion of 15% - 20% unchanged drug, at 500mg two times a day, urinary excretion increases to about 36%. The major urinary metabolite is 14-hydroxycarithromycin and accounts for 10% - 15% of the dose. The majority of the remainder of the dose is eliminated in faeces, mainly via bile, with 5% - 10% clarithromycin being recovered in faeces.

Clarithromycin plasma concentrations are increased with 500mg three times a day compared to 500mg two times a day.

The tissue concentrations of clarithromycin are several times higher than circulating plasma levels, increased levels have been found in tonsil and lung tissue. At therapeutic levels it is 80% plasma protein bound. It penetrates the gastric mucus, the levels in gastric tissue and mucus are higher with co-administration of omeprazole than without.

It penetrates the middle ear fluid, with levels being higher than in serum.

5.3. Preclinical Safety Data

Clarithromycin toxicity was related to dose and duration of treatment, with the primary target organ being the liver in all species, hepatic lesions occurring after 14 days in dog and monkey. Toxic mg/kg doses were higher than dose recommended for patients.

A range of *in vitro* and *in vivo* tests did not demonstrate mutagenic potential for clarithromycin. Fertility and reproduction animal studies have not shown adverse effects. Some animal studies have suggested an embryotoxic effect although at levels that were toxic to the mother.

6. Pharmaceutical Particulars

6.1. List of Excipients

Croscarmellose, cellulose microcrystalline, colloidal anhydrous silica, povidone, stearic acid, magnesium stearate, talc, hypromellose, propylene glycol, sorbitan oleate, vanilla, quinoline yellow (E104), titanium dioxide (E171) hydroxy propyl cellulose, and sorbic acid.

6.2. Incompatibilities

None known

6.3. Shelf Life

36 months when stored at a temperature below 30°C

6.4. Special Precautions for Storage

Store below 30°C in the original package

6.5. Nature and Contents of Container

Tablets are packed into PVC/PVDC-Al blisters.

KLERIMED tablets 250mg: Boxes of 14 and 100 tablets

KLERIMED tablets 500mg: Boxes of 14, 98 and 700 tablets

6.6. Instruction for Use/Handling

None

Administrative Data**7. Manufacturer**

MEDOCHEMIE (Far East) LTD, (Oral Facility), Np. 40, VSIP II Street, Vietnam-Singapore Industrial Park II, Binh Duong Industry-Service-Urban Complex, Hoa Phu ward, Thu Dau Mot city, Binh Duong province, Vietnam

8. Product Registration Holder

KOMEDIC SDN BHD, 4, Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor D.E., Malaysia

9. Product Registration Number

KLERIMED tablets 250 mg: MAL05101667AZ

KLERIMED tablets 500 mg: MAL05101668AZ

10. Date of (Partial) Revision of the Text: September 2021